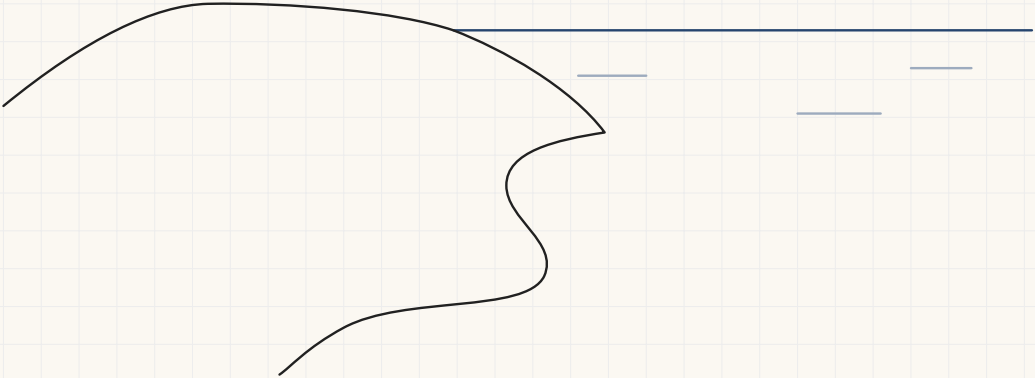
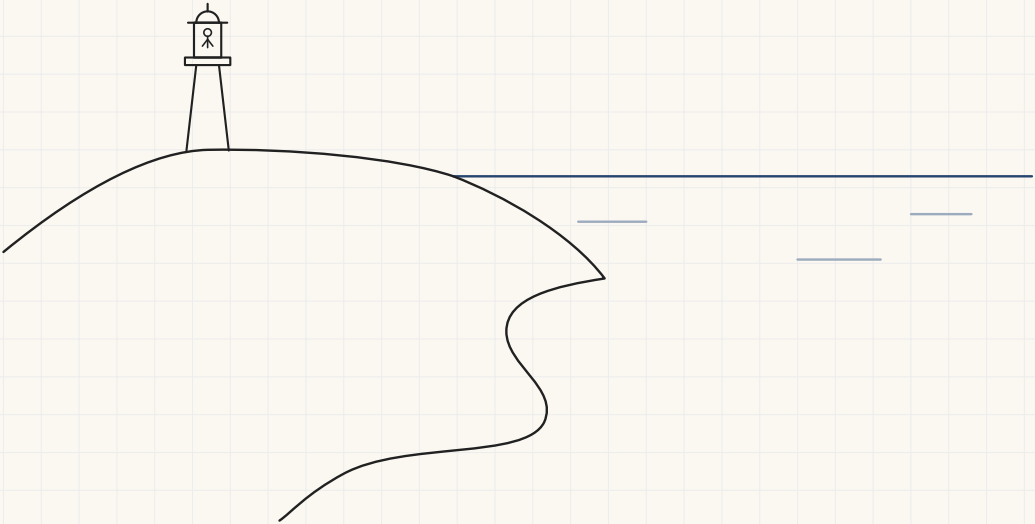
Manifolds

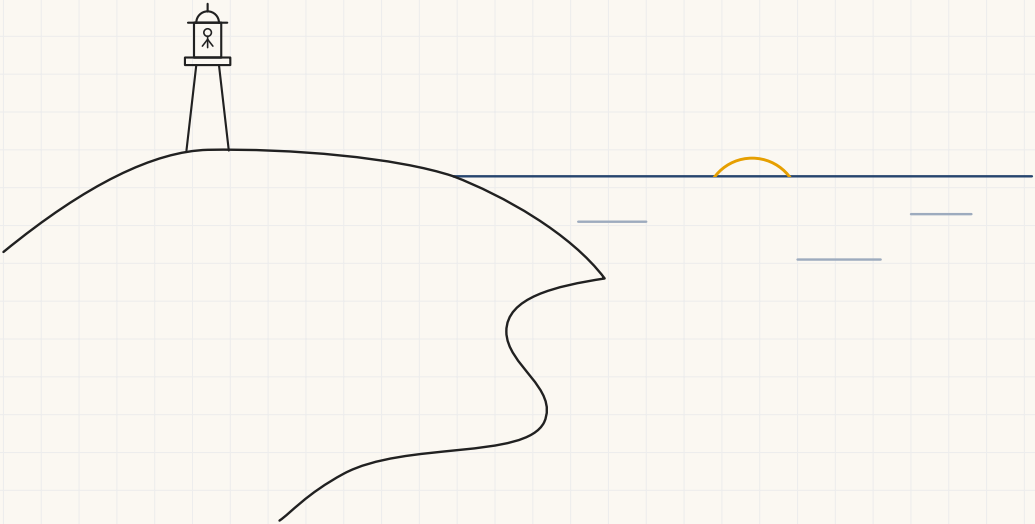
Weekend 4, Friday morning

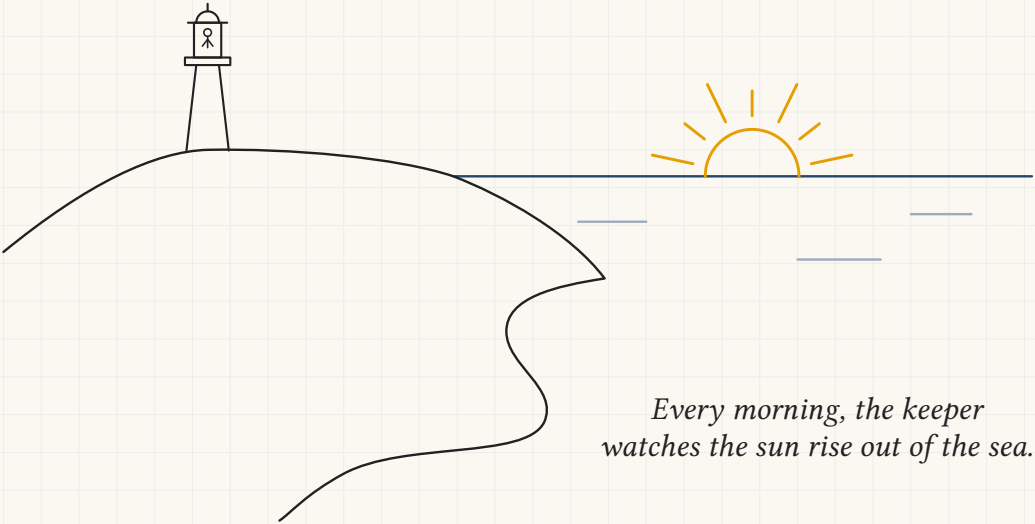
Carlos Cotrini

Friday 23 October 2026

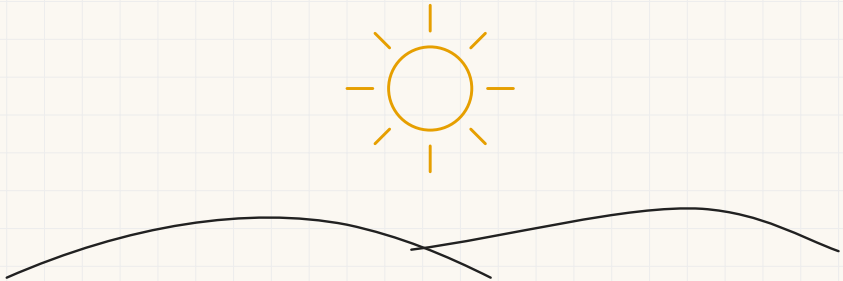


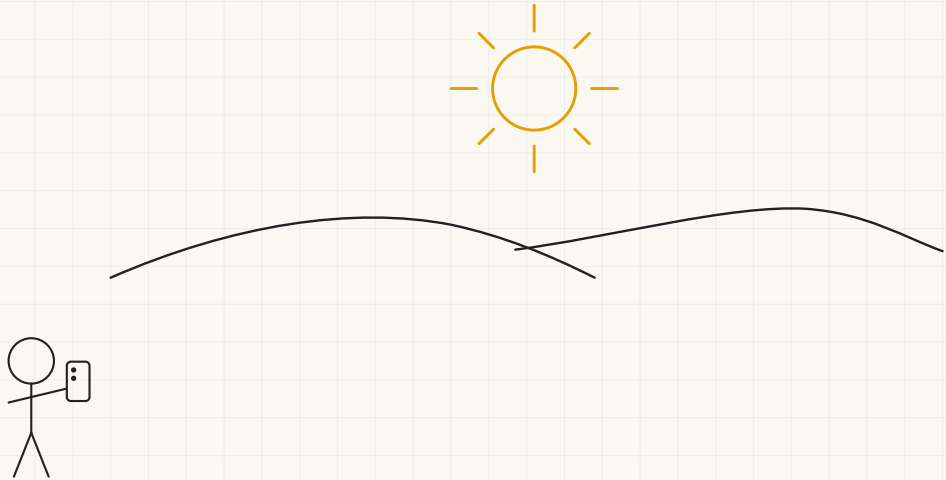


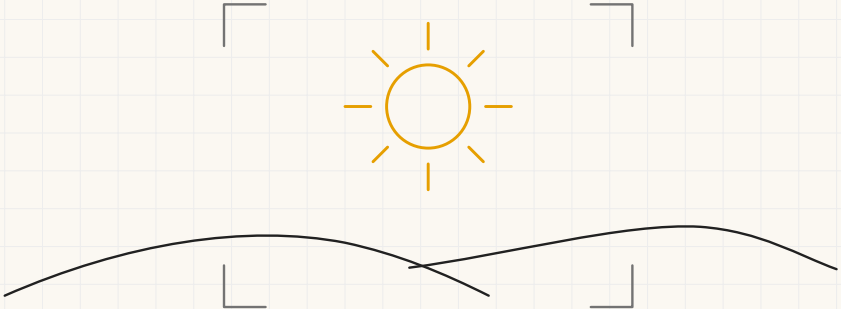




Every morning, the keeper watches the sun rise out of the sea.



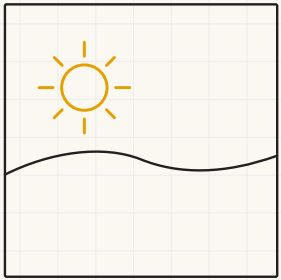




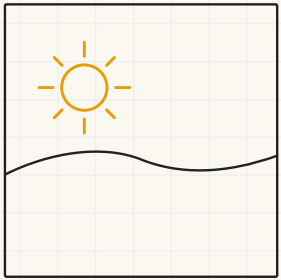


Inland, someone holds up a phone and keeps the moment:
a sunrise, saved as millions of pixels.

Predicting from a photo



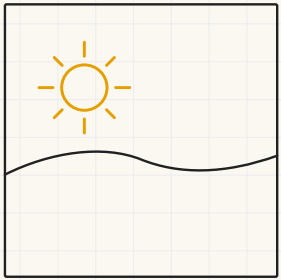
Predicting from a photo



Regression

Time of the day

Predicting from a photo



Regression

Time of the day

Classification

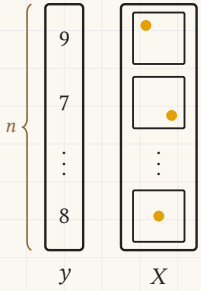
Season

1

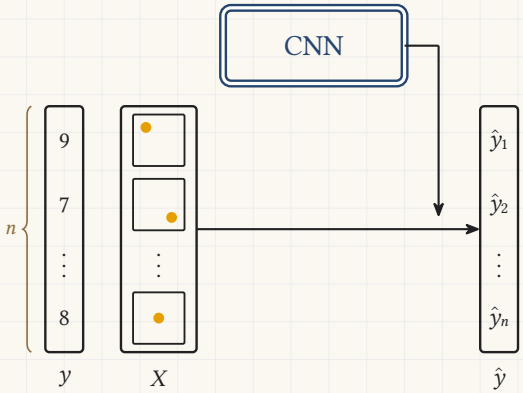
Section 1

From pixels to coordinates

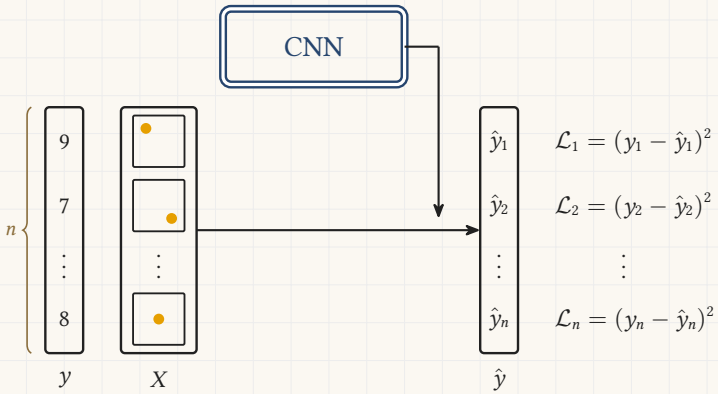
Vapnik's statistical learning



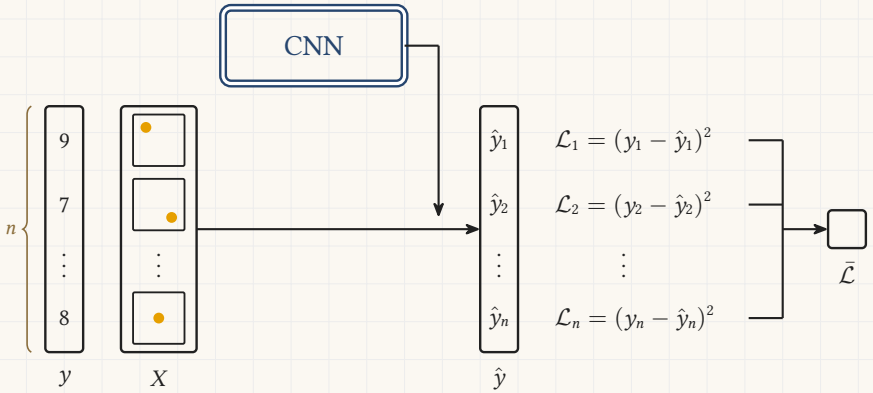
Vapnik's statistical learning



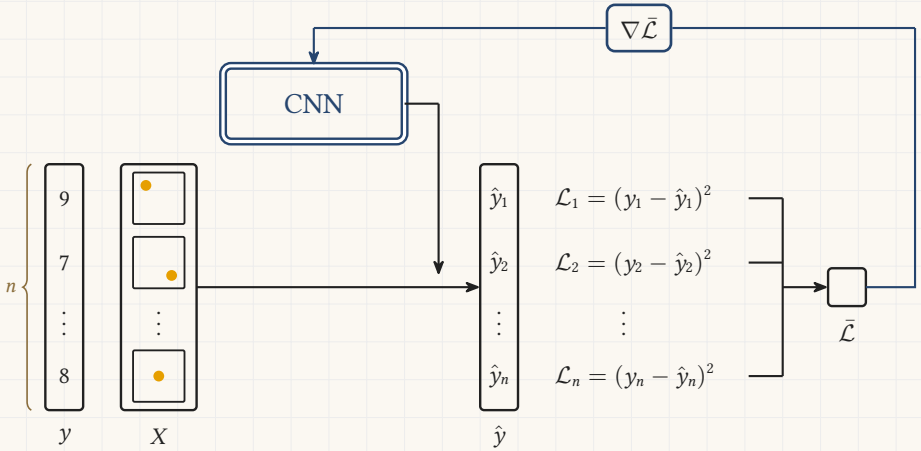
Vapnik's statistical learning



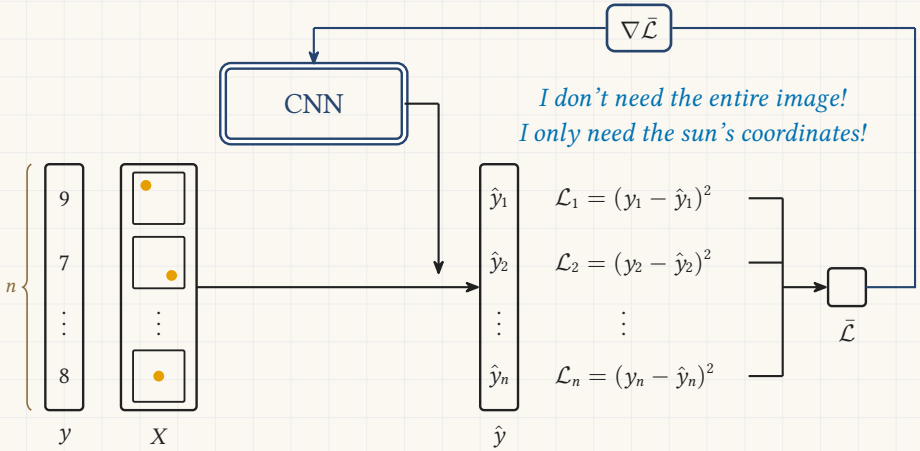
Vapnik's statistical learning



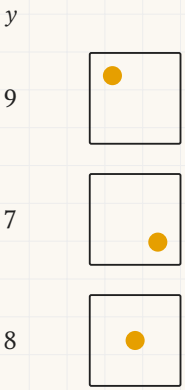
Vapnik's statistical learning



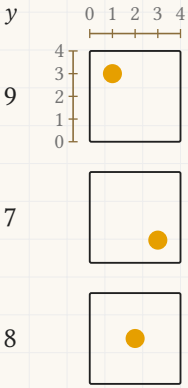
Vapnik's statistical learning



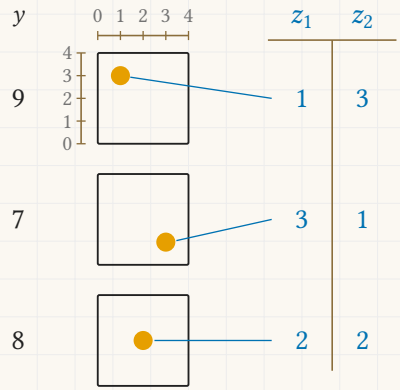
The sun’s coordinates



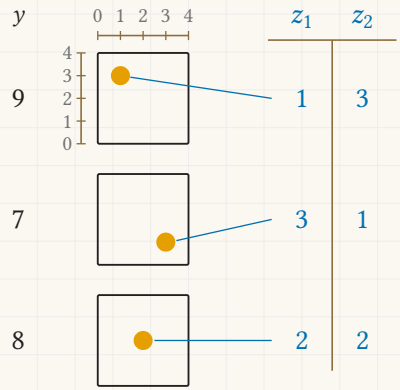
The sun's coordinates



The sun's coordinates

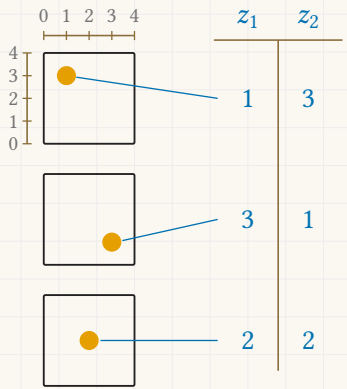


The sun's coordinates



$$\hat{y} = w\,z_1 + b$$

The sun's coordinates



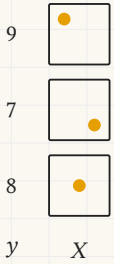
$$\hat{y} = w\,z_1 + b$$

$$\hat{y}_1 = w \cdot 1 + b$$

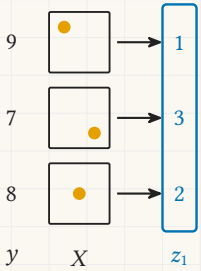
$$\hat{y}_2 = w \cdot 3 + b$$

$$\hat{y}_3 = w \cdot 2 + b$$

Learning w and b

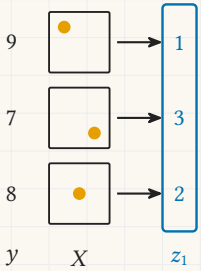


Learning w and b

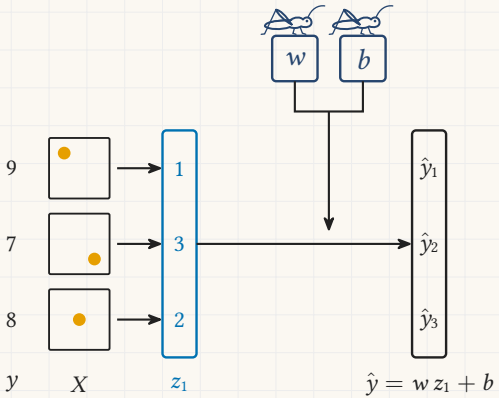


Learning w and b

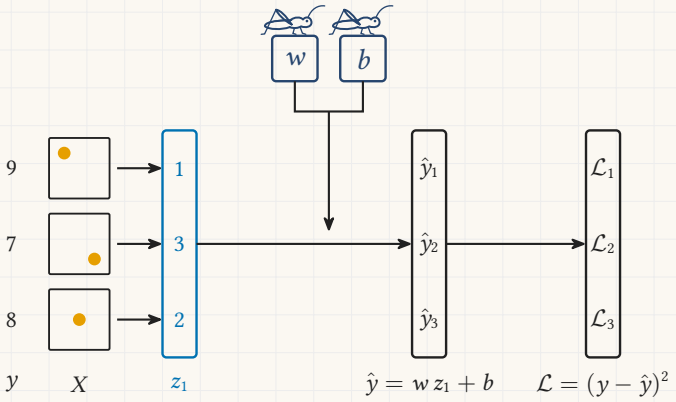




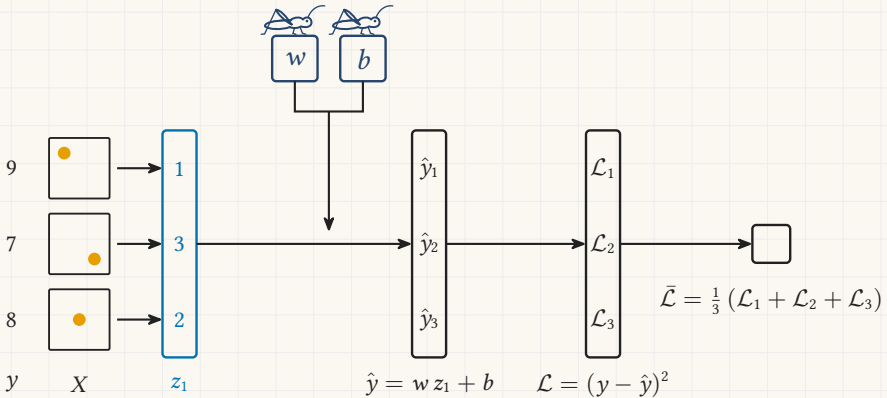
Learning w and b



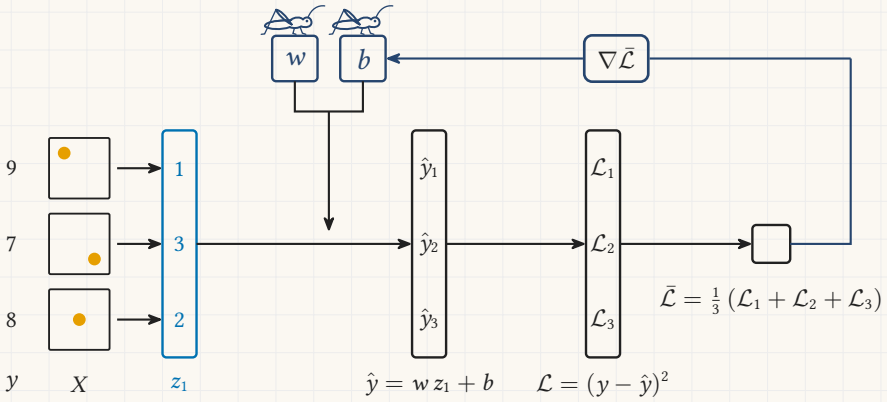
Learning w and b



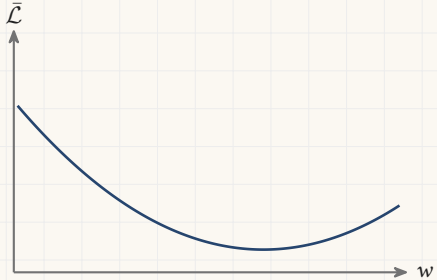
Learning w and b

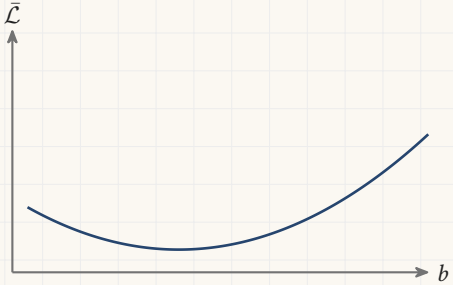


Learning w and b

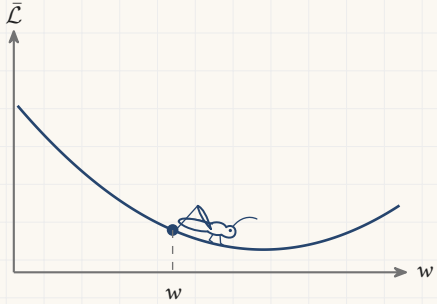


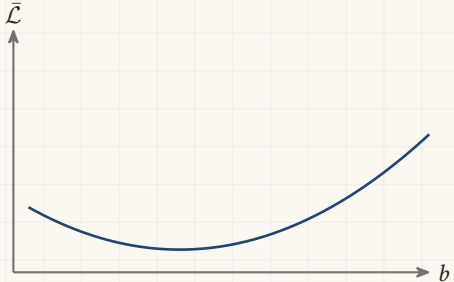
Each grasshopper measures its slope



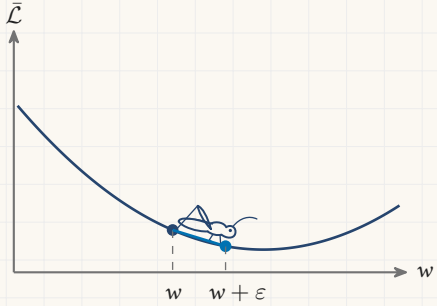


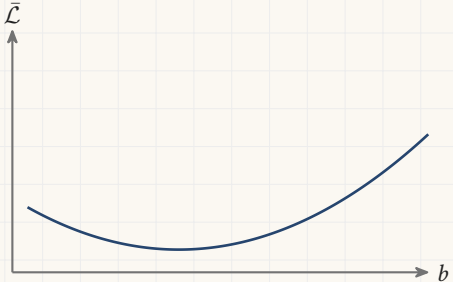
Each grasshopper measures its slope



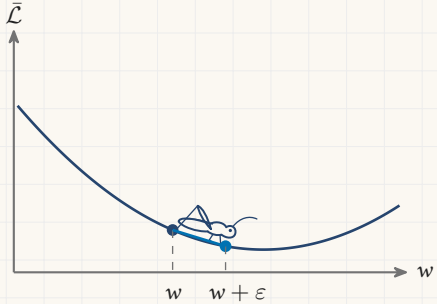


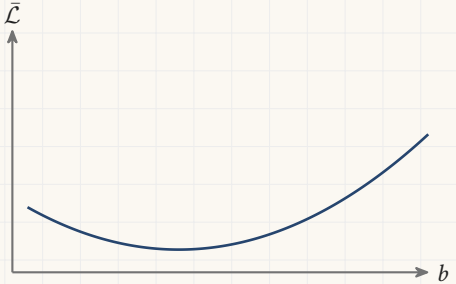
Each grasshopper measures its slope





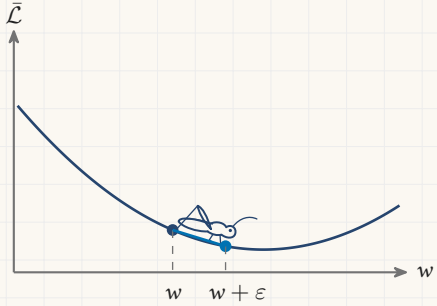
Each grasshopper measures its slope

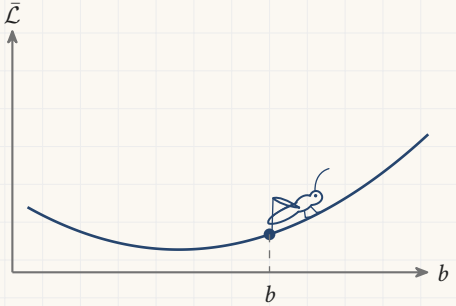




$$\frac{\partial \bar{\mathcal{L}}}{\partial w} \approx \frac{\bar{\mathcal{L}}(w + \varepsilon,\ b) - \bar{\mathcal{L}}(w,\ b)}{\varepsilon}$$

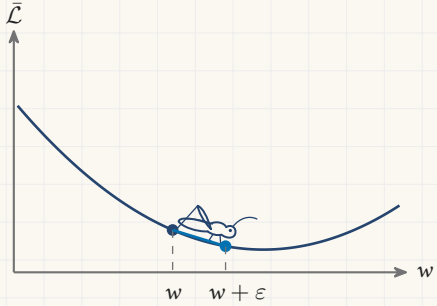
Each grasshopper measures its slope

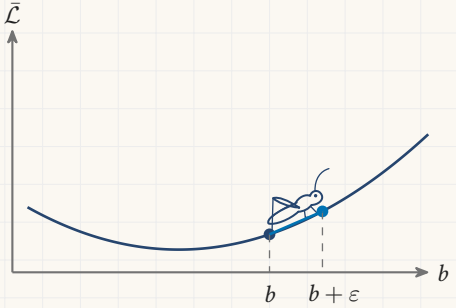




$$\frac{\partial \bar{\mathcal{L}}}{\partial w} \approx \frac{\bar{\mathcal{L}}(w+\varepsilon,\ b) - \bar{\mathcal{L}}(w,\ b)}{\varepsilon}$$

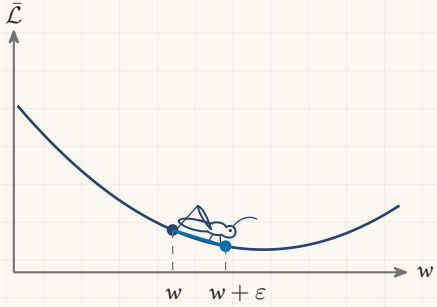
Each grasshopper measures its slope



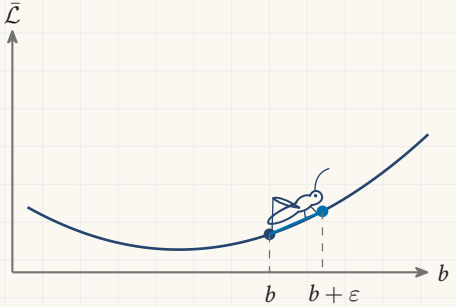


$$\frac{\partial \bar{\mathcal{L}}}{\partial w} \approx \frac{\bar{\mathcal{L}}(w+\varepsilon,\ b) - \bar{\mathcal{L}}(w,\ b)}{\varepsilon}$$

Each grasshopper measures its slope



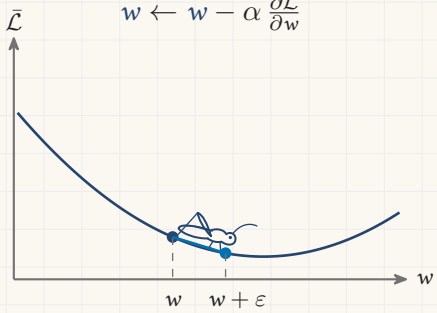
$$\frac{\partial \bar{\mathcal{L}}}{\partial w} \approx \frac{\bar{\mathcal{L}}(w+\varepsilon,\ b) - \bar{\mathcal{L}}(w,\ b)}{\varepsilon}$$



$$\frac{\partial \bar{\mathcal{L}}}{\partial b} \approx \frac{\bar{\mathcal{L}}(w,\ b+\varepsilon) - \bar{\mathcal{L}}(w,\ b)}{\varepsilon}$$

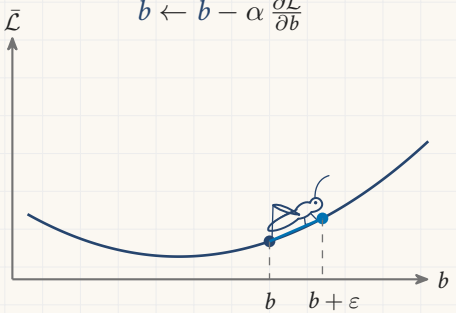
Each grasshopper measures its slope

$$w \leftarrow w - \alpha \frac{\partial \bar{\mathcal{L}}}{\partial w}$$



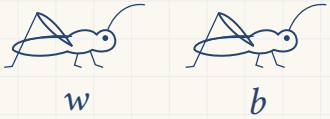
$$\frac{\partial \bar{\mathcal{L}}}{\partial w} \approx \frac{\bar{\mathcal{L}}(w + \varepsilon,\ b) - \bar{\mathcal{L}}(w,\ b)}{\varepsilon}$$

$$b \leftarrow b - \alpha \frac{\partial \bar{\mathcal{L}}}{\partial b}$$



$$\frac{\partial \bar{\mathcal{L}}}{\partial b} \approx \frac{\bar{\mathcal{L}}(w,\ b + \varepsilon) - \bar{\mathcal{L}}(w,\ b)}{\varepsilon}$$

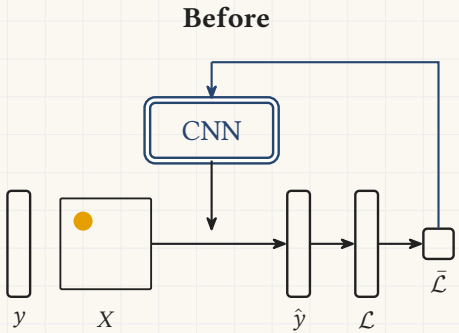
The grasshoppers, click by click



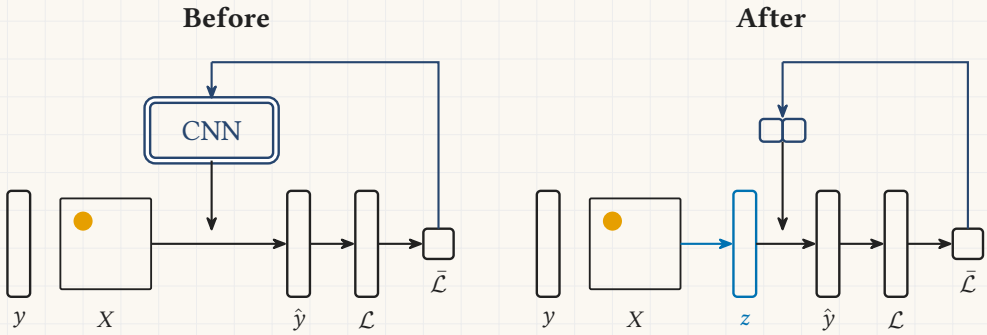
Type where the two grasshoppers start, then follow them, one click at a time.

https://eth-bmai-hs26.github.io/BMAI-PAGE/viz/we4/grasshopper-regression/

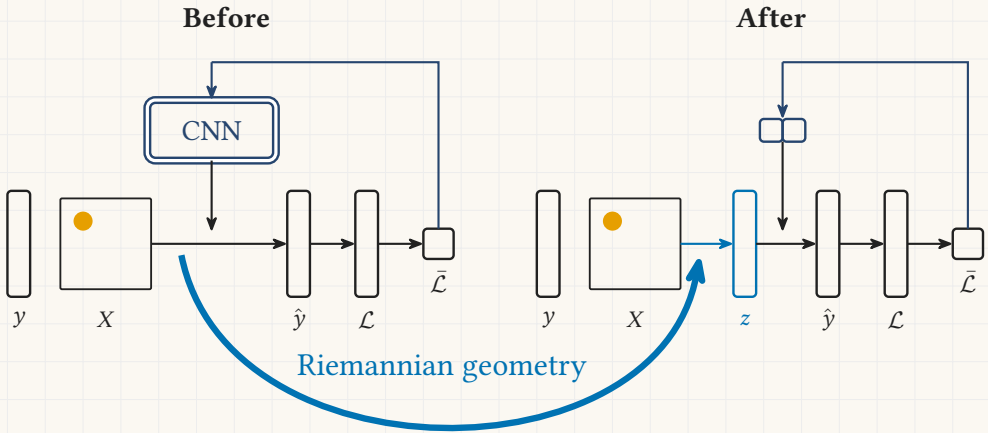
Before and after



Before and after



Before and after

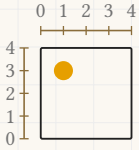


2

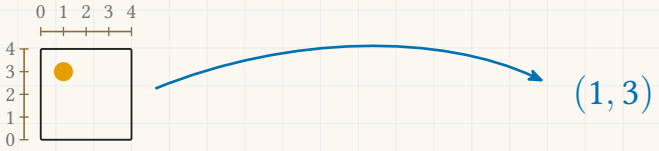
Section 2

Charts

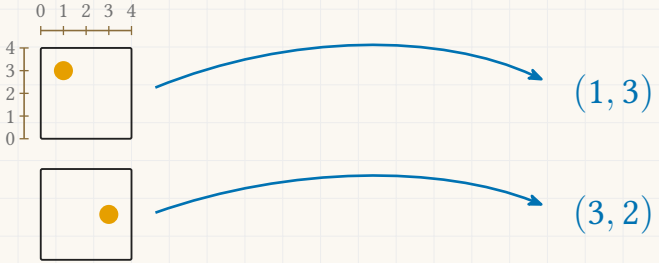
From the photo to its coordinates, and back



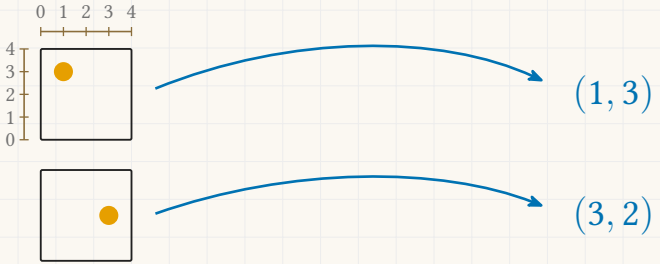
From the photo to its coordinates, and back



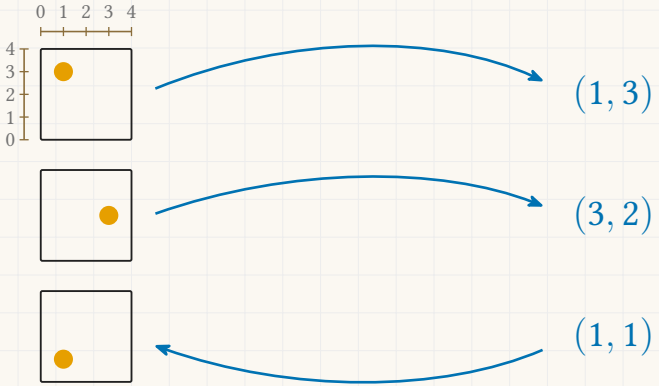
From the photo to its coordinates, and back



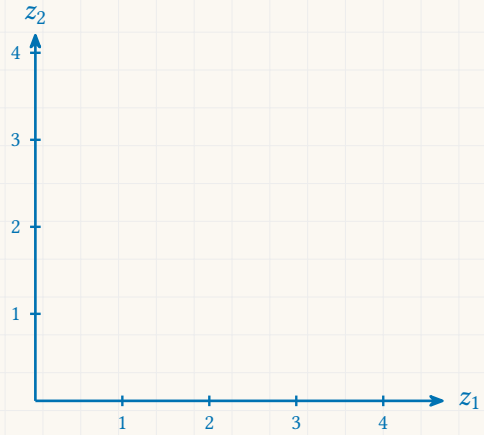
From the photo to its coordinates, and back



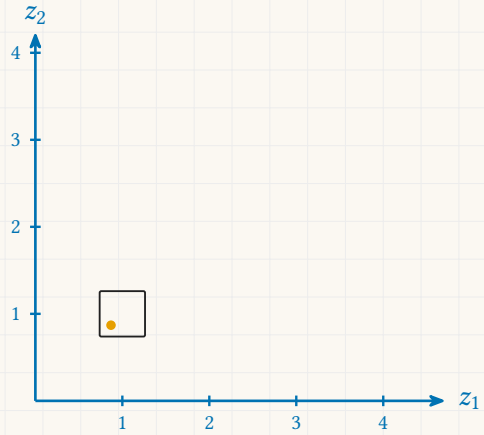
From the photo to its coordinates, and back



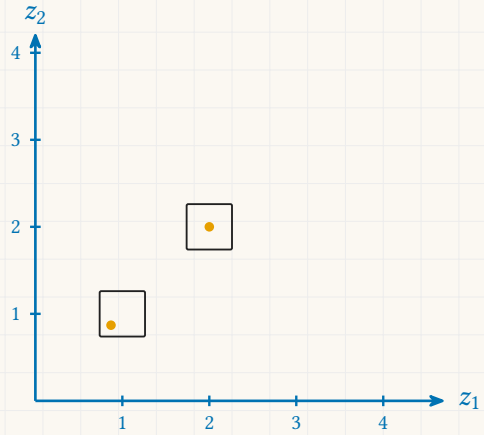
The photos, placed at their coordinates



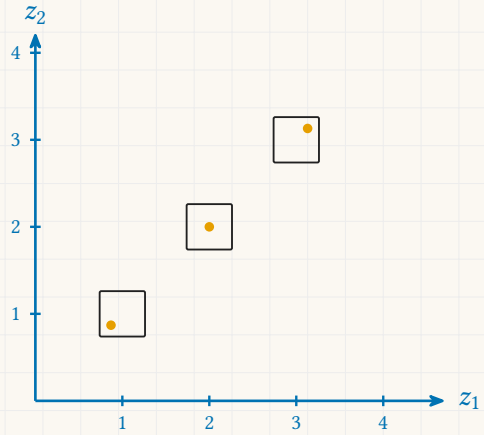
The photos, placed at their coordinates



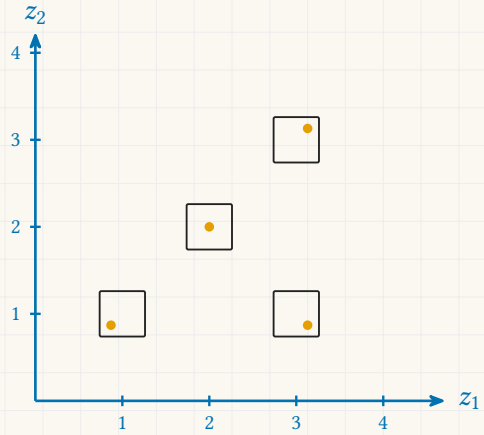
The photos, placed at their coordinates



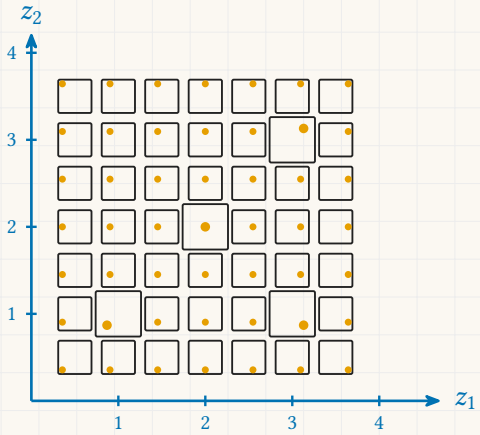
The photos, placed at their coordinates



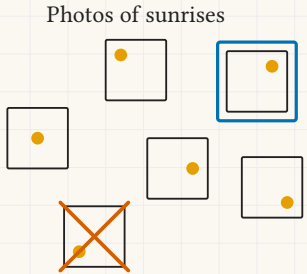
The photos, placed at their coordinates



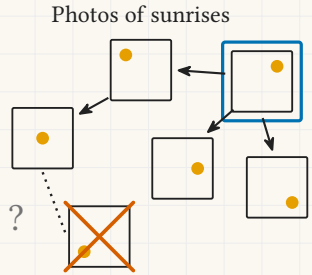
The photos, placed at their coordinates



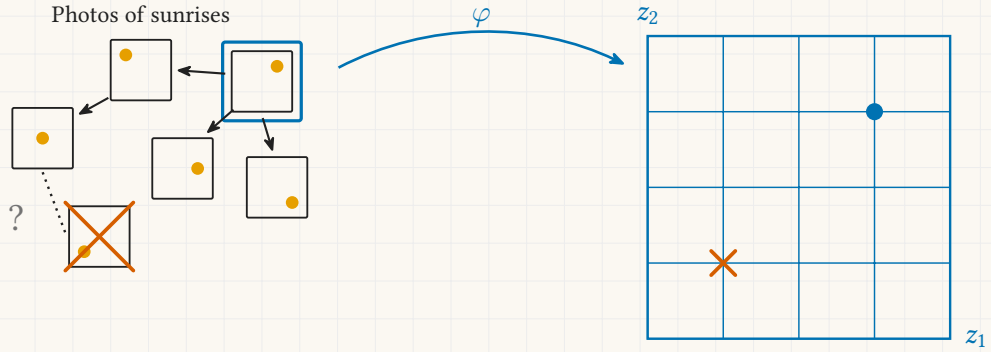
Walking through the sunrises



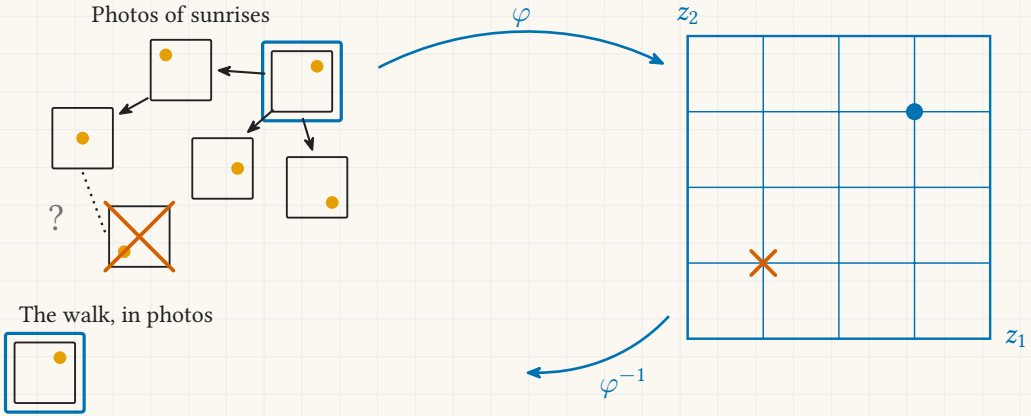
Walking through the sunrises



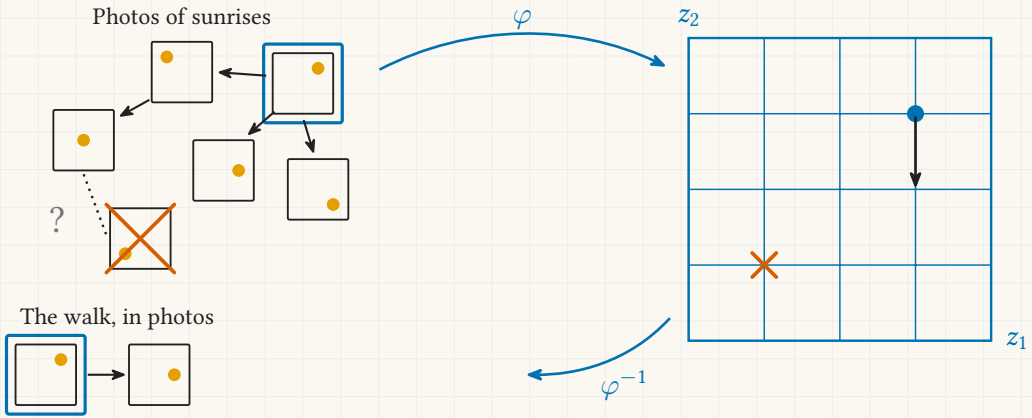
Walking through the sunrises



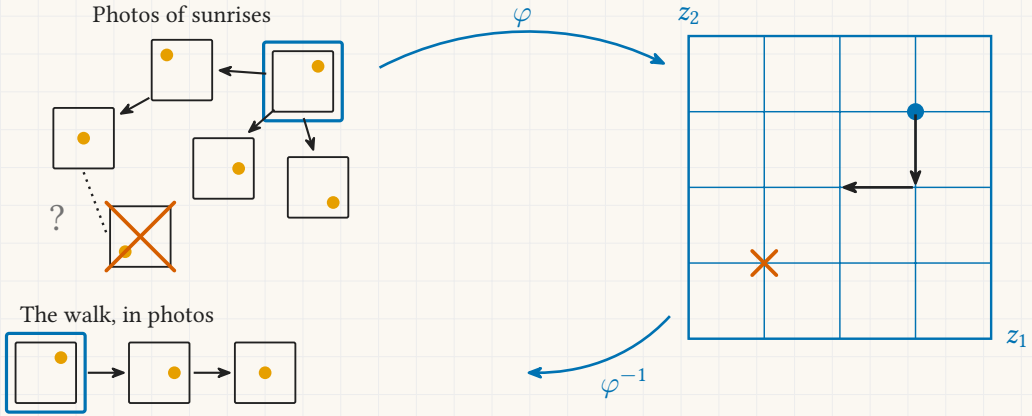
Walking through the sunrises



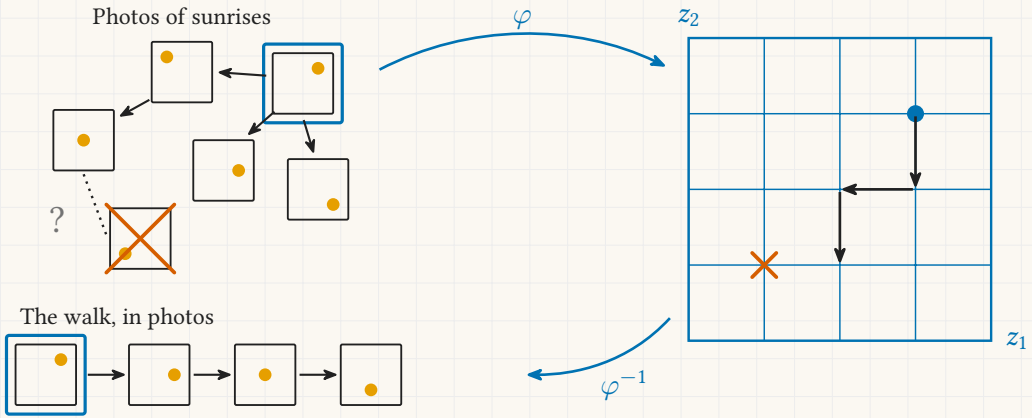
Walking through the sunrises



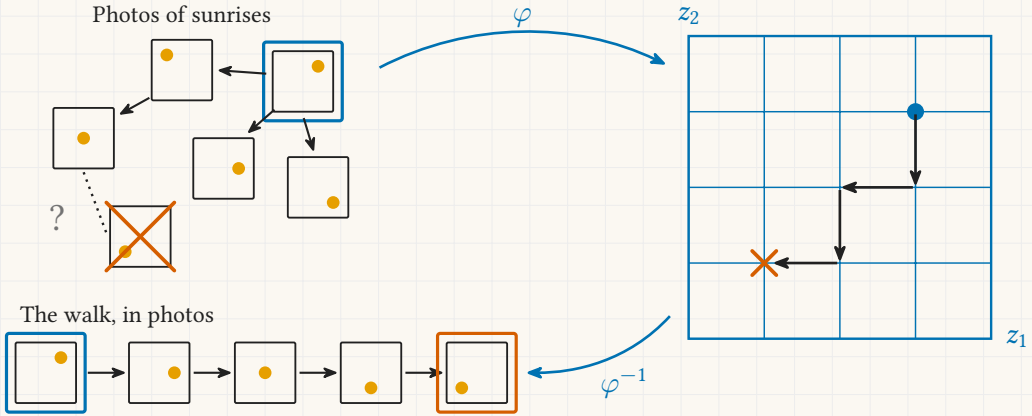
Walking through the sunrises



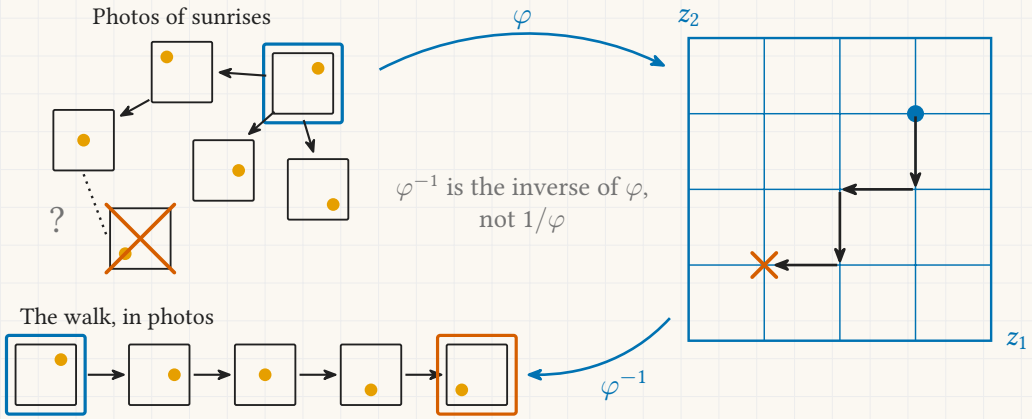
Walking through the sunrises



Walking through the sunrises



Walking through the sunrises



Manifolds

The set of all sunrises forms a manifold.

Manifolds

The set of all sunrises forms a manifold.

A manifold is any set of objects that admits a homeomorphism to a navigation chart.*

* A white lie: in general, a manifold only needs such a chart around each of its points.

Another example



Another example





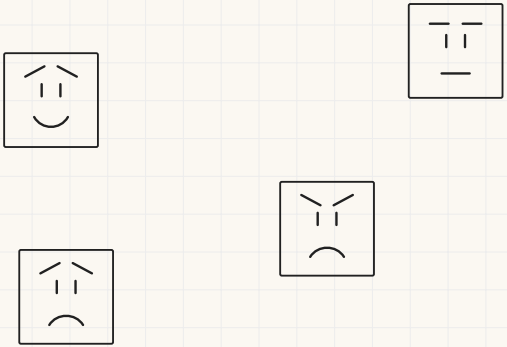
Another example



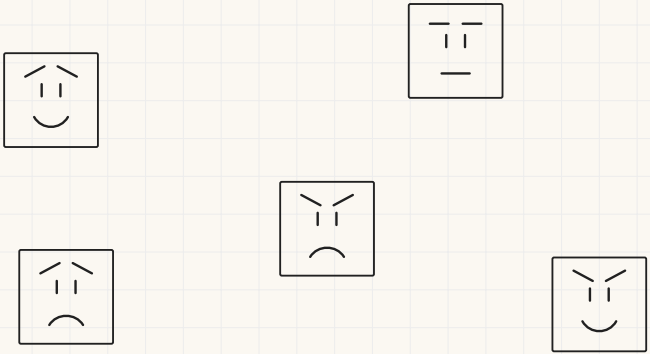




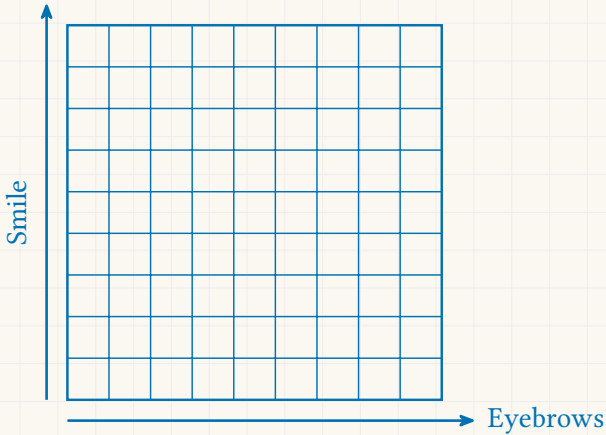
Another example



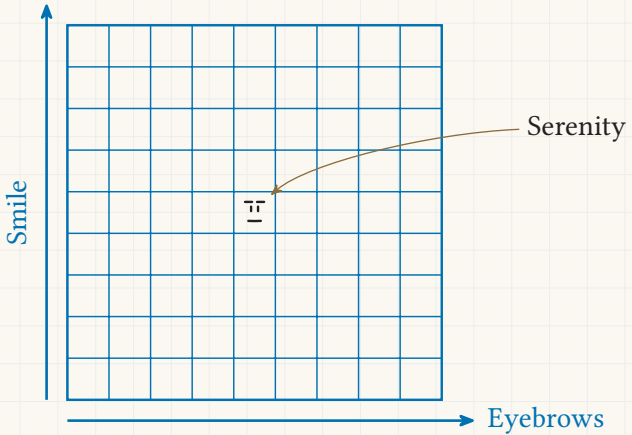
Another example



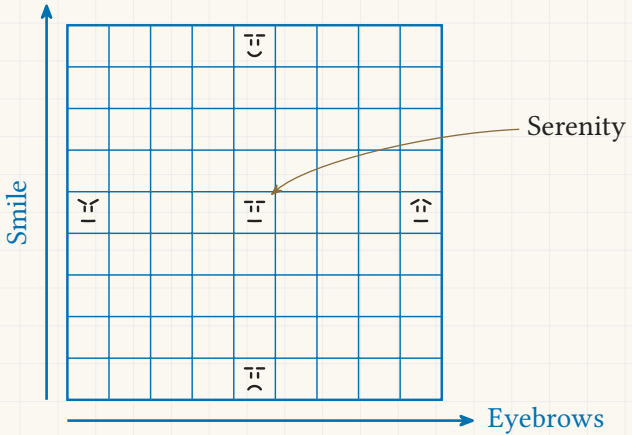
A chart of faces



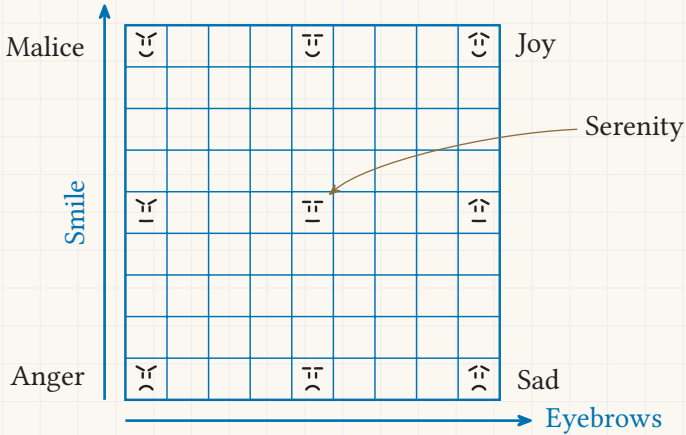
A chart of faces



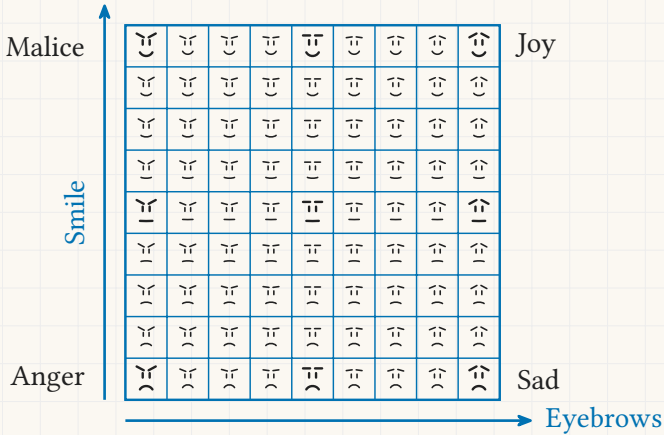
A chart of faces



A chart of faces



A chart of faces

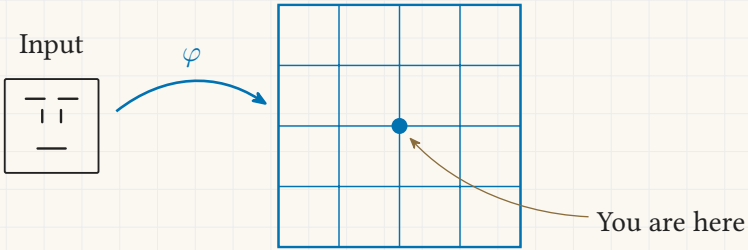


The main application of Riemannian geometry to AI

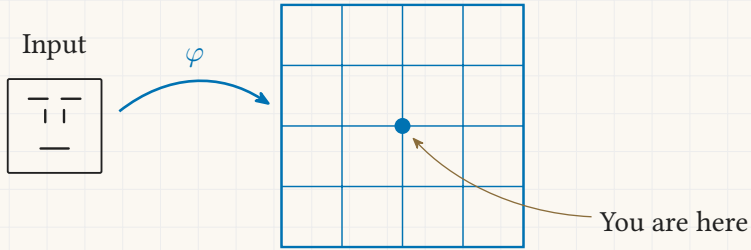
Input



The main application of Riemannian geometry to AI



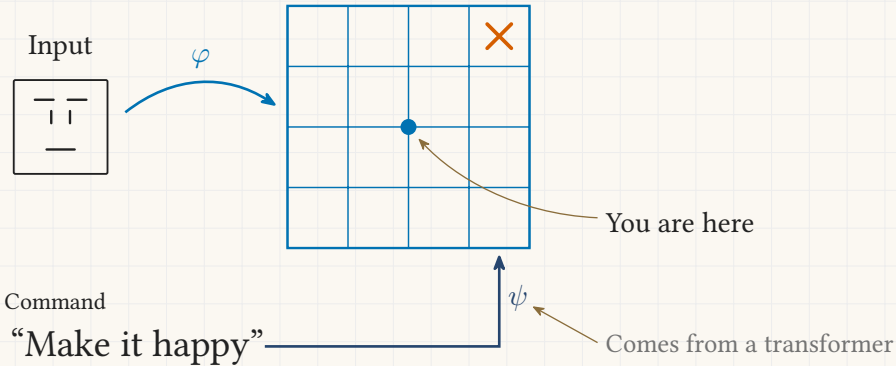
The main application of Riemannian geometry to AI



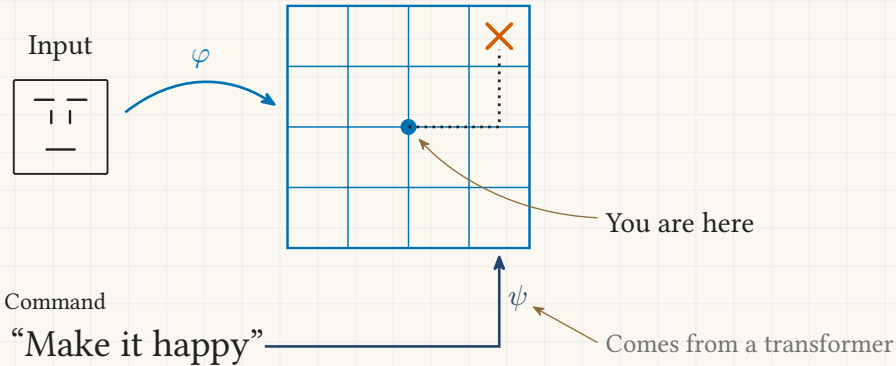
Command

“Make it happy”

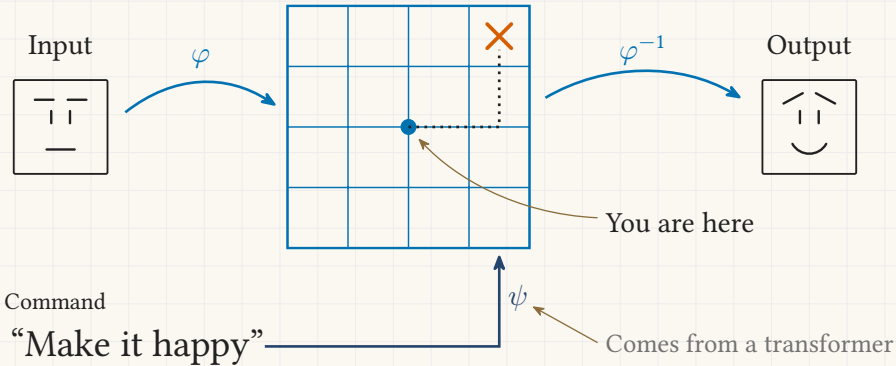
The main application of Riemannian geometry to AI



The main application of Riemannian geometry to AI



The main application of Riemannian geometry to AI

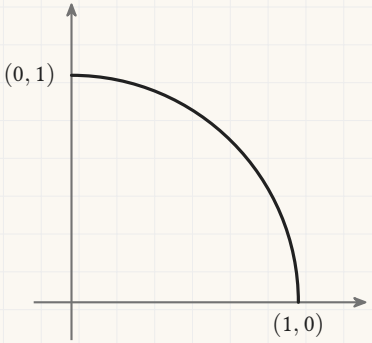


3

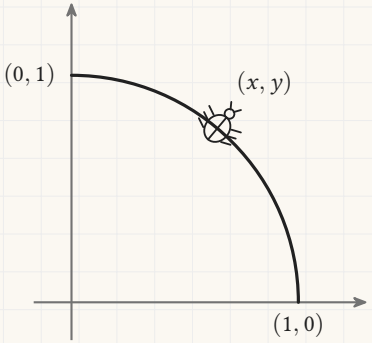
Section 3

Examples of manifolds

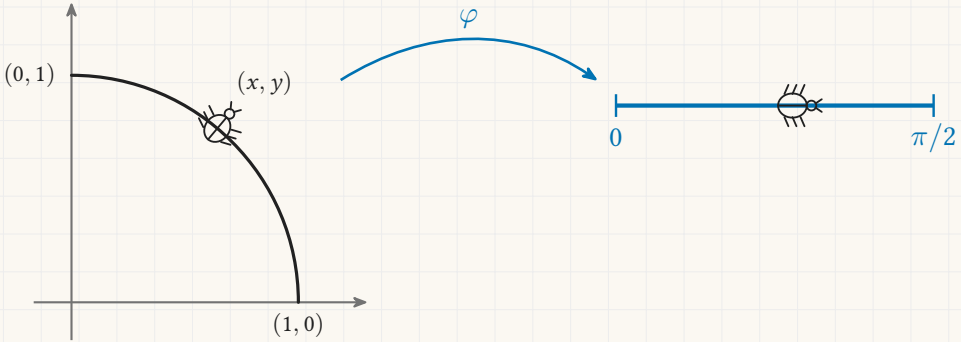
An arc of the circle



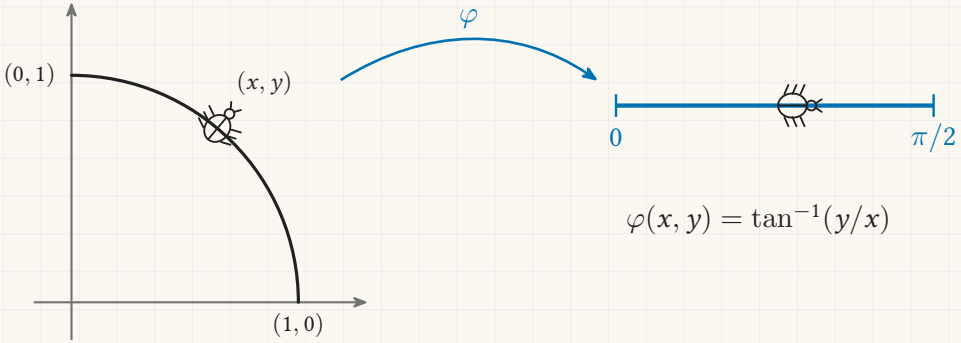
An arc of the circle



An arc of the circle

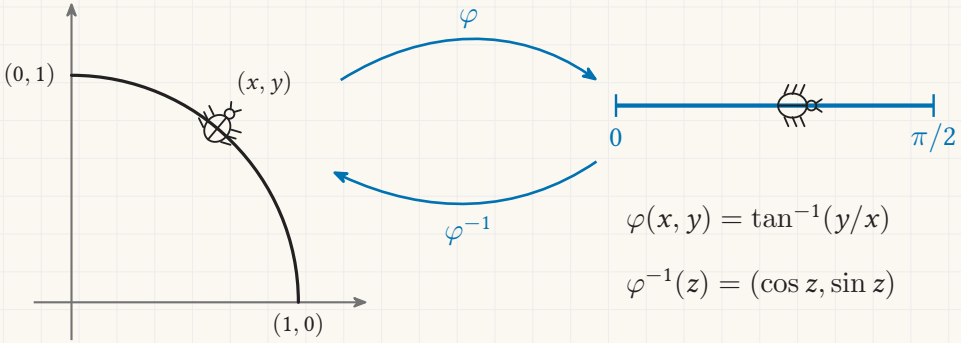


An arc of the circle



$$\varphi(x,y) = \tan^{-1}(y/x)$$

An arc of the circle

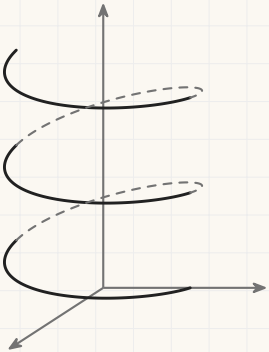


$$\varphi(x, y) = \tan^{-1}(y/x)$$

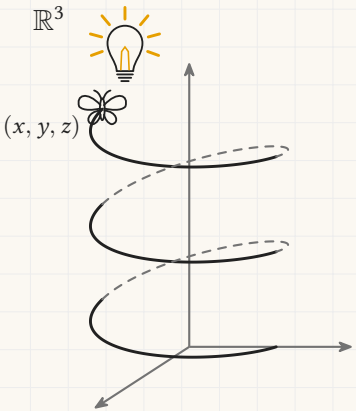
$$\varphi^{-1}(z) = (\cos z, \sin z)$$

A helix

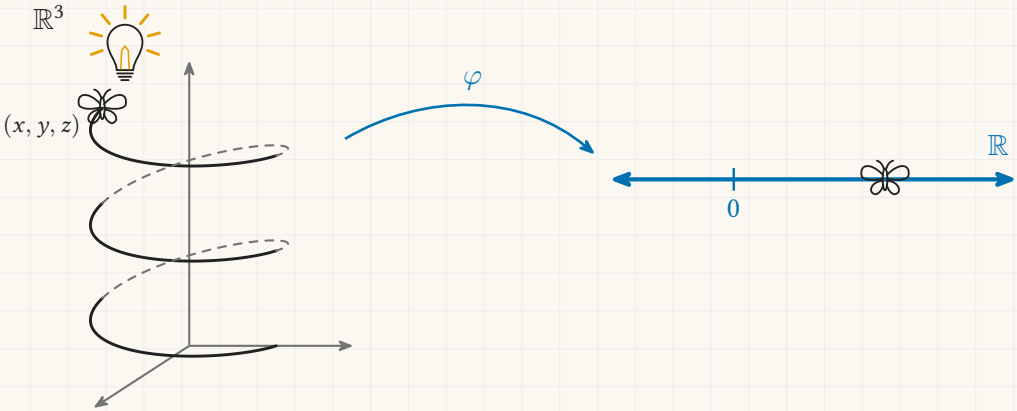
$\mathbb{R}^3$



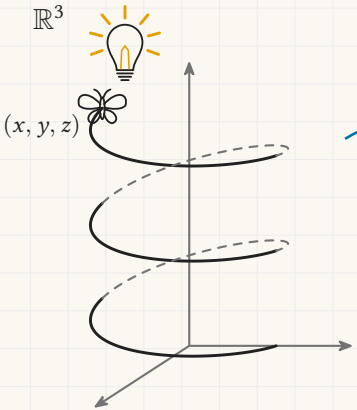
A helix



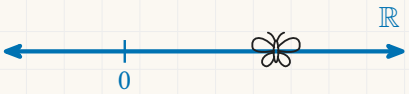
A helix



A helix



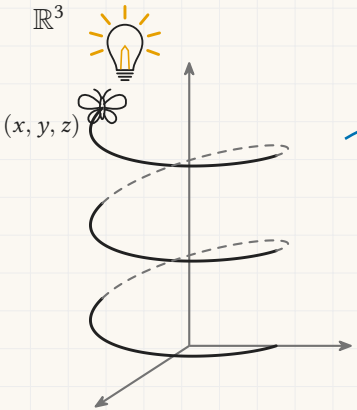
φ

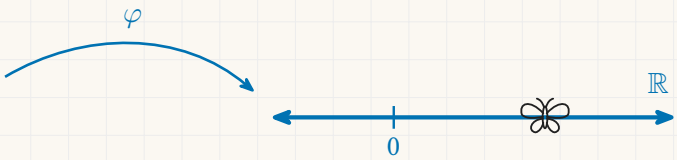


$\varphi^{-1}(t) =$

$\varphi(x, y, z) =$

A helix

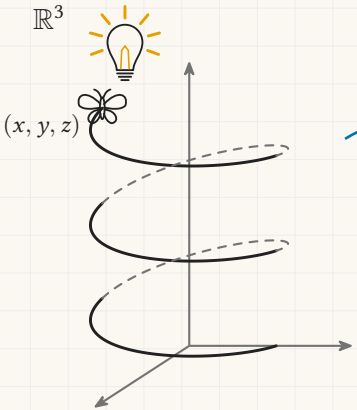




$$\varphi^{-1}(t) = (\cos t,\ \sin t,\ t)$$

$$\varphi(x, y, z) =$$

A helix







$$\varphi^{-1}(t) = (\cos t,\ \sin t,\ t)$$

$$\varphi(x, y, z) = z$$

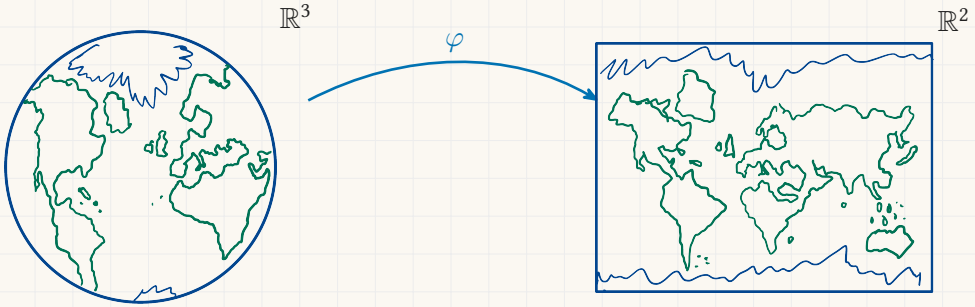
The globe and the world map

$\mathbb{R}^3$

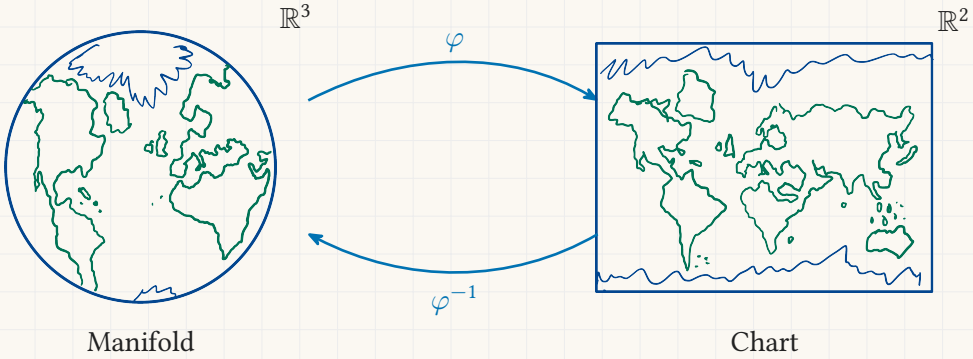


Manifold

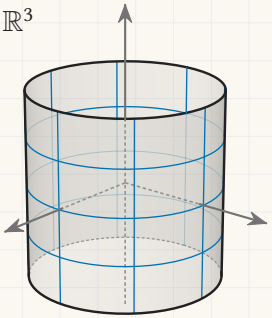
The globe and the world map



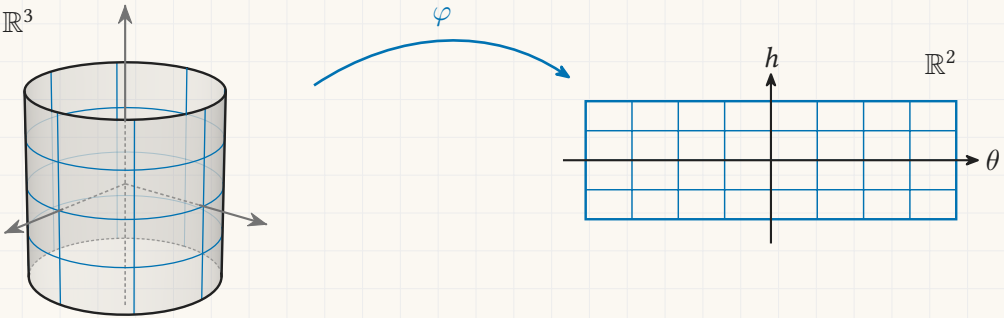
The globe and the world map



A cylinder

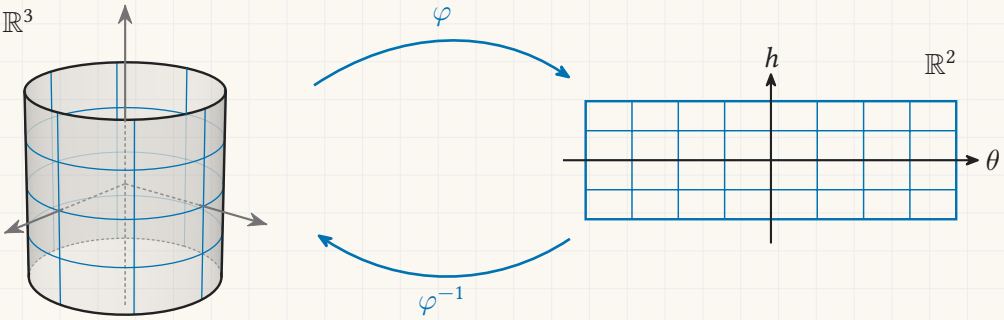


A cylinder

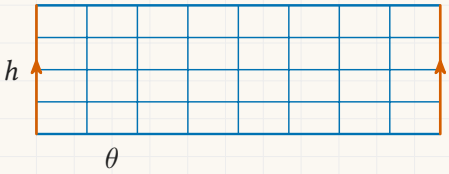


A cylinder

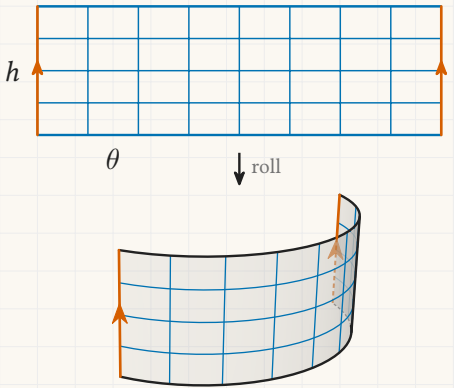
$$\varphi^{-1}(\theta, h) = \left(\cos \tfrac{2\pi\theta}{360},\ \sin \tfrac{2\pi\theta}{360},\ h\right)$$



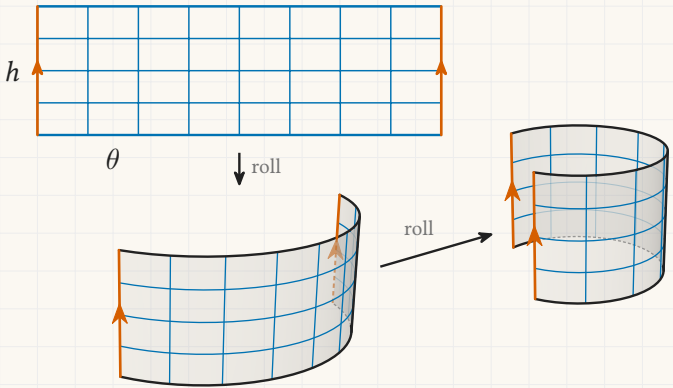
Folding a chart into a cylinder



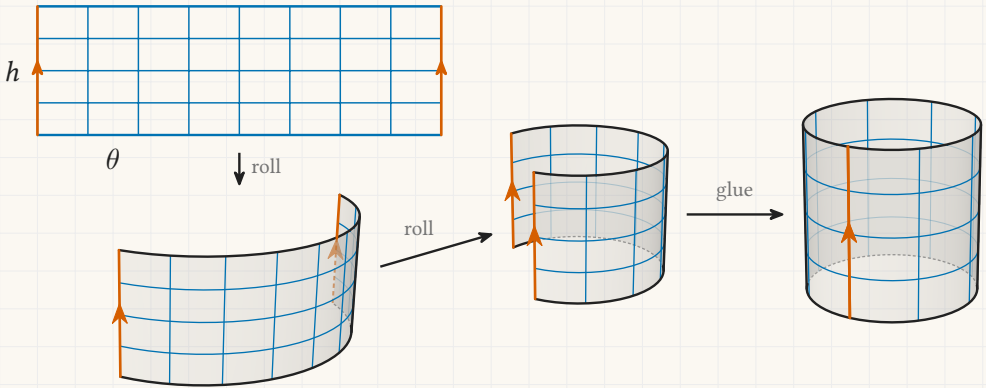
Folding a chart into a cylinder



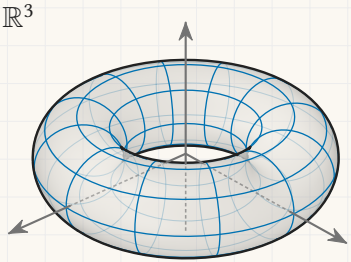
Folding a chart into a cylinder



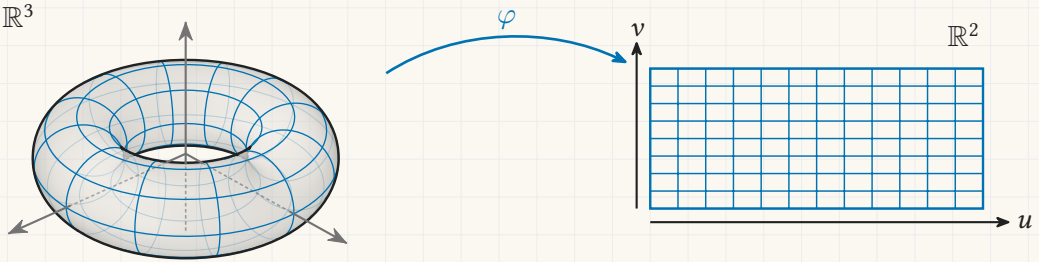
Folding a chart into a cylinder



A torus



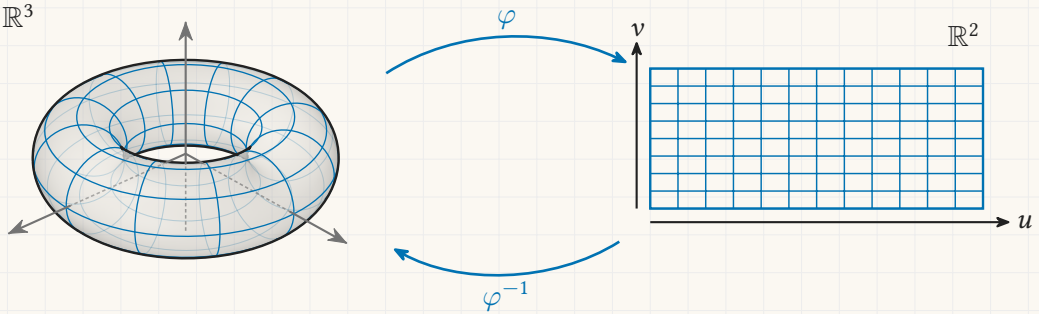
A torus



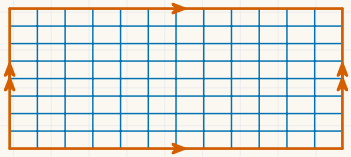
A torus

$$\varphi^{-1}(u, v) = \big((R + r\cos v)\cos u,\ (R + r\cos v)\sin u,\ r\sin v\big)$$

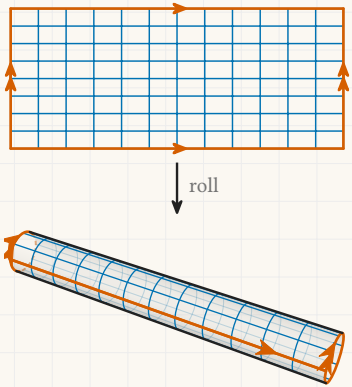
R: centre to the middle of the tube. r: radius of the tube.



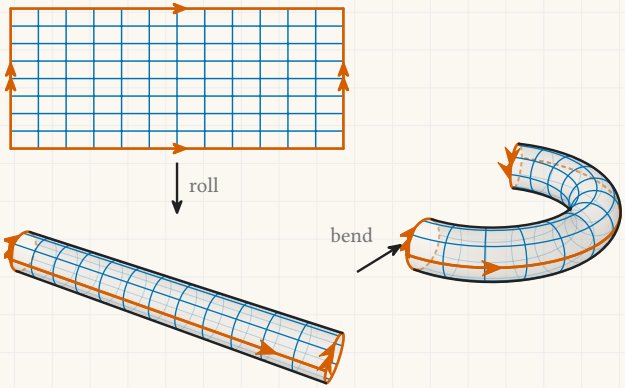
Folding a chart into a torus



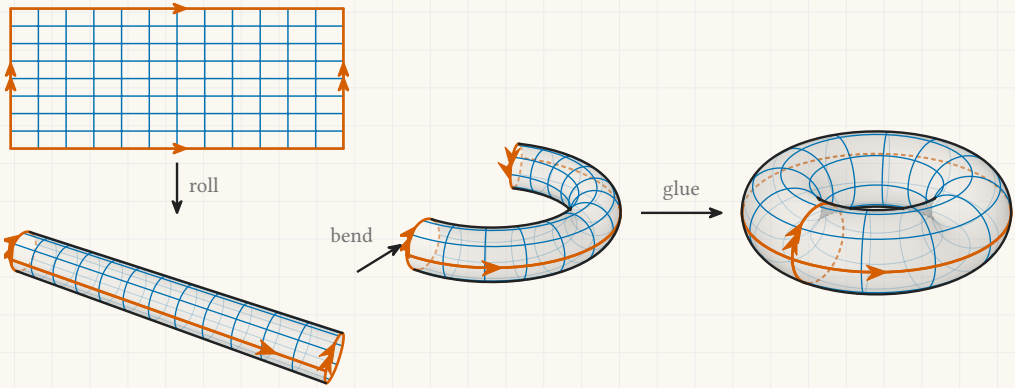
Folding a chart into a torus



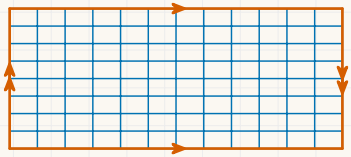
Folding a chart into a torus



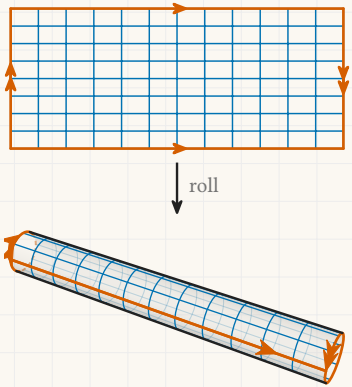
Folding a chart into a torus



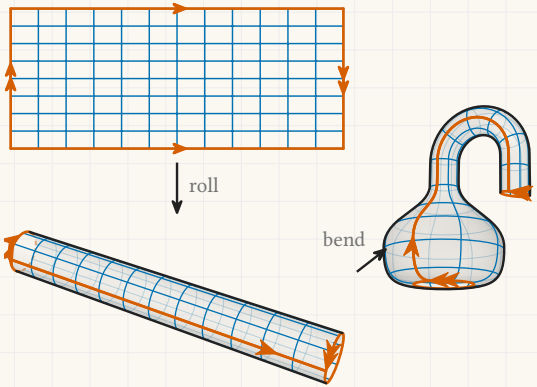
Folding a chart into a Klein bottle



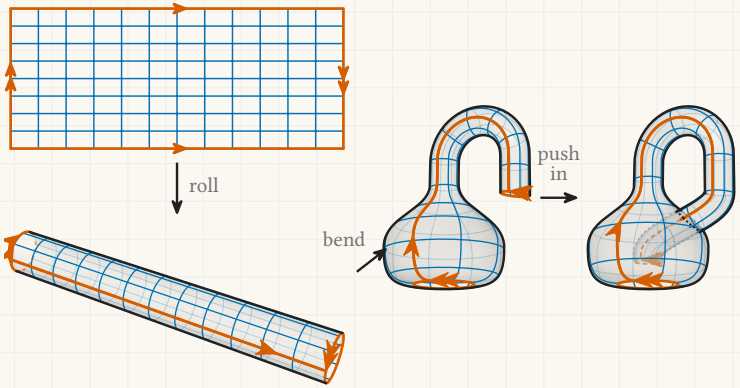
Folding a chart into a Klein bottle



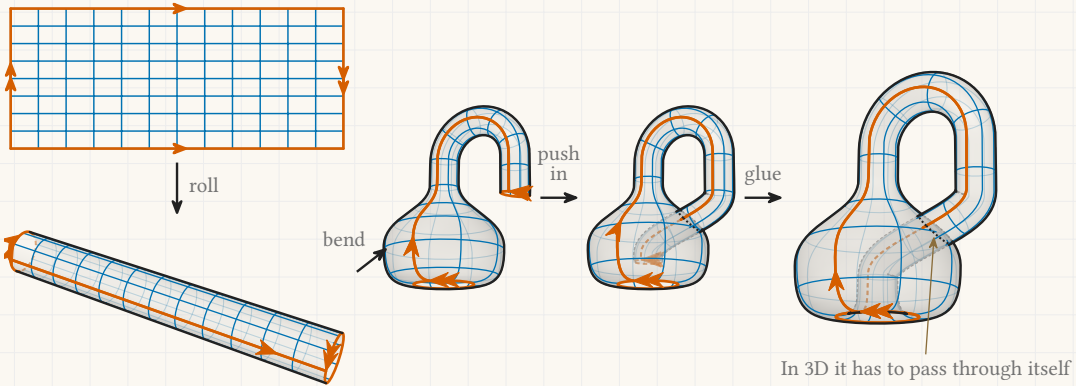
Folding a chart into a Klein bottle



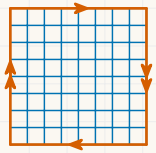
Folding a chart into a Klein bottle



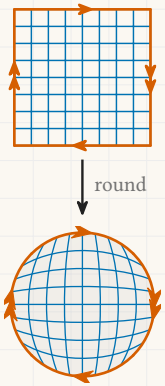
Folding a chart into a Klein bottle



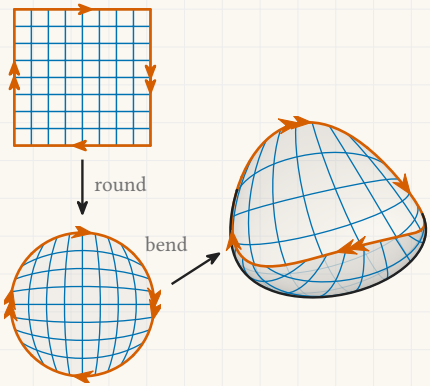
Folding a chart into a projective plane



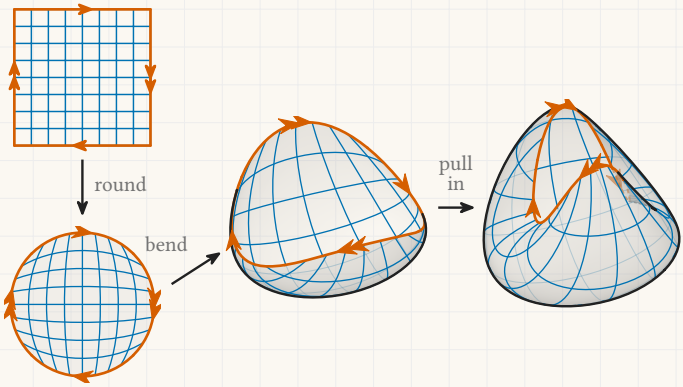
Folding a chart into a projective plane



Folding a chart into a projective plane



Folding a chart into a projective plane



Folding a chart into a projective plane

